



Faculty of Engineering & Technology

Electrical & Computer Engineering Department

DIGITAL LABORATORY - ENCS2110

**Experiment 3: Encoders, Decoders, Multiplexer and
Demultiplexer**

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Abstract:

Our goal in this experiment to understand the principles and be familiar of different combinational circuits (encoders/decoders, multiplexers/demultiplexers), and be with knowledge of how they work and how and when we should use them in the best effective way, by using them to implement functions, and having more complex circuits, which do more than simple things, in order to use them in our projects in future.

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Introduction; -

A combinational circuit is the digital logic circuit in which the output depends on the combination of inputs at that point of time with total disregard to the past state of the inputs. The digital logic gate is the building block of combinational circuits. The function implemented by combinational circuit is depend upon the Boolean expressions.

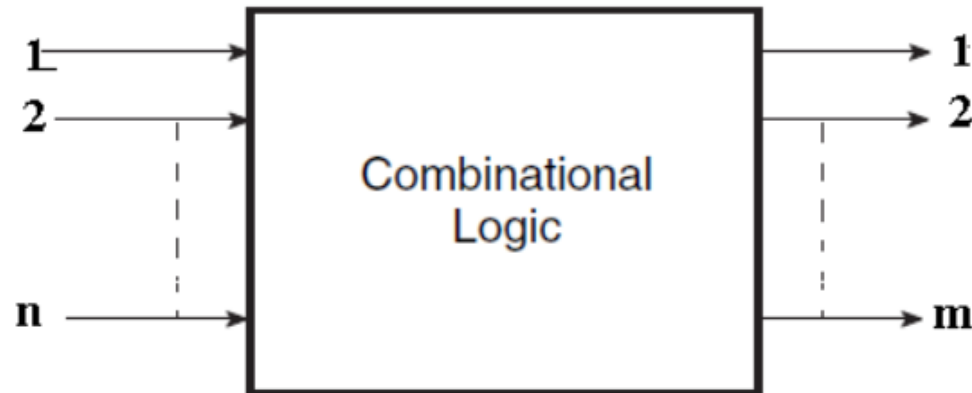


Figure 1: Combinational Circuit

theory

Decoders: -

The combinational circuit that changes the binary information into 2^N output lines is known as Decoders. The binary information is passed in the form of N input lines. The output lines define the 2^N - bit code for the binary information. In simple words, the Decoder performs the reverse operation of the Encoder. At a time, only one input line is activated for simplicity. The produced 2^N -bit output code is equivalent to the binary information. The outputs of the decoder are nothing, but the min terms of ' N ' input variables lines as shown in figure 2.

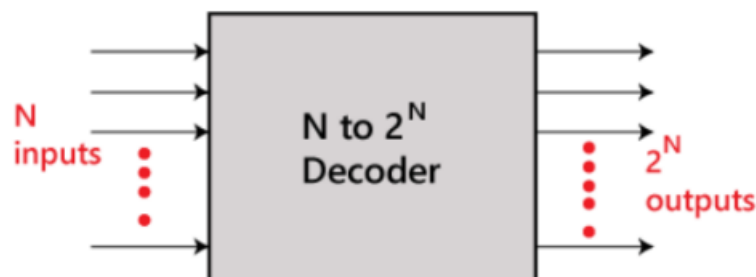


Figure 2: General Decoder block diagram.

Enable	Inputs		Outputs			
E	A1	A0	Y3	Y2	Y1	Y0
0	x	x	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

Figure 3: Truth table of 2 to 4 decoder

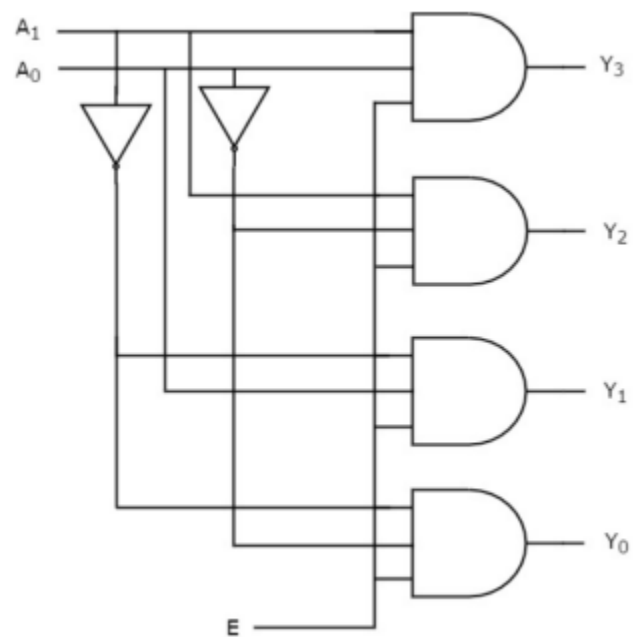


Figure 4: 2 to 4 Decoder circuit.

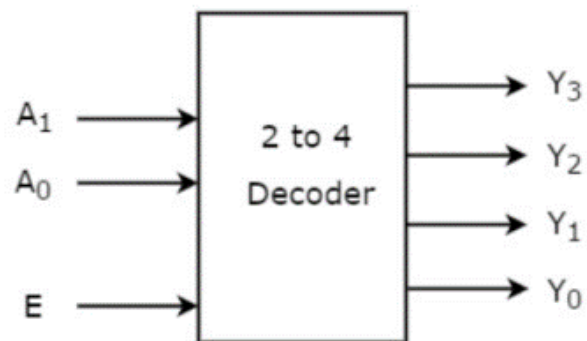


Figure 5: 2 to 4 Decoder with enable

Encoders:

The combinational circuits that change the binary information into N output lines are known as **Encoders**. The binary information is passed in the form of 2^N input lines. The output lines define the N -bit code for the binary information. In simple words, the **Encoder** performs the reverse operation of the **Decoder**. At a time, only one input line is activated for simplicity. The produced N -bit output code is equivalent to the binary information.

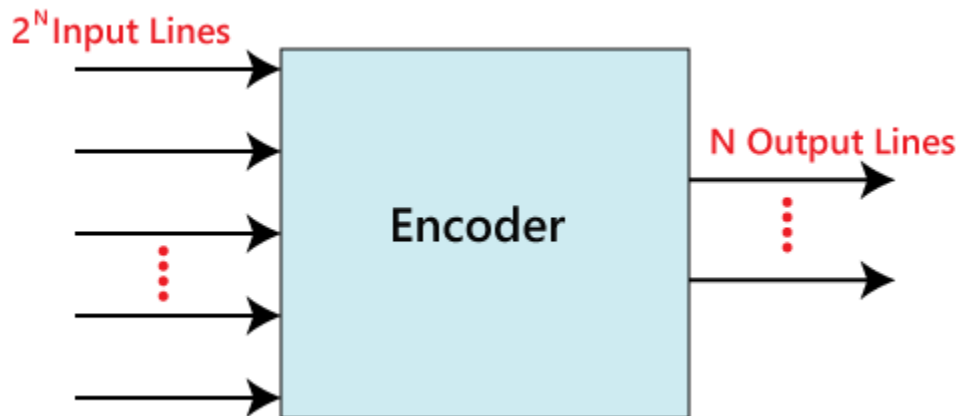


Figure 6:Encoders

Example: - 4-to-2-line Encoder:

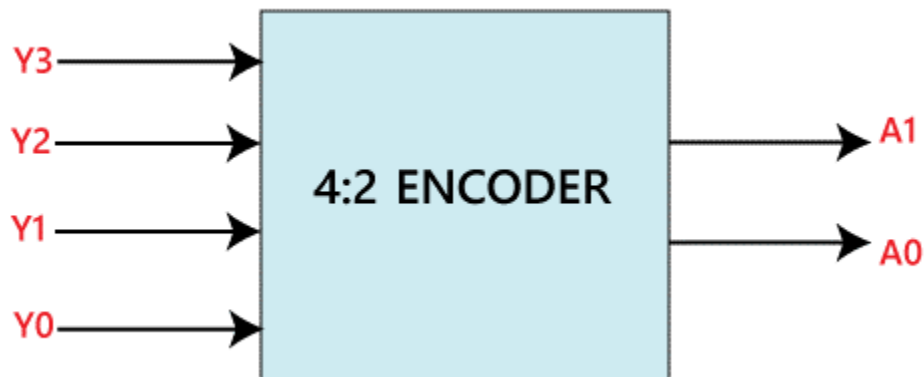


Figure 7:Block Diagram:

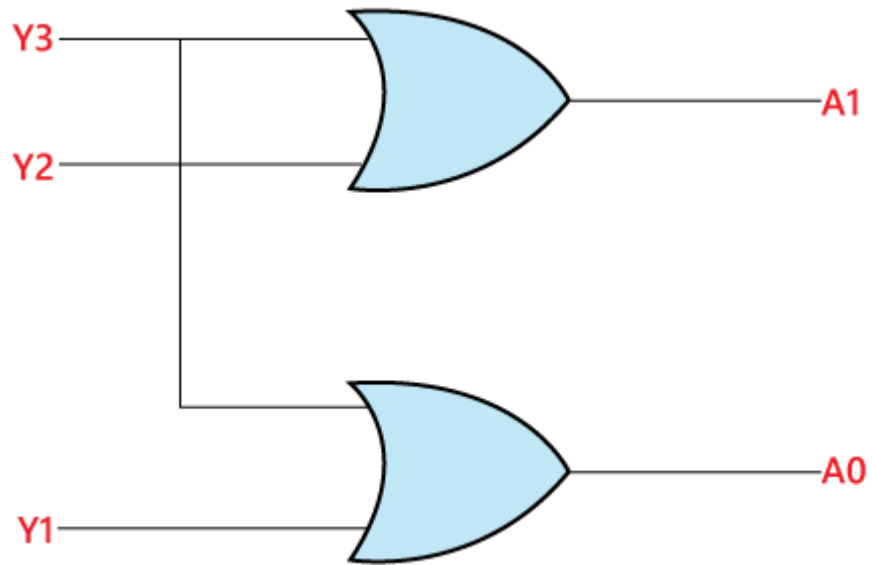


Figure 8: Logical circuit of the 4x2 Encoder

INPUTS				OUTPUTS	
Y_3	Y_2	Y_1	Y_0	A_1	A_0
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

Figure 9: Truth Table of 4x2 Encoder

Multiplexer:

A multiplexer is a combinational circuit that has 2^n input lines and a single output line. Simply, the multiplexer is a multi-input and single-output combinational circuit. The binary information is received from the input lines and directed to the output line. Based on the values of the selection lines, one of these data inputs will be connected to the output.

Unlike encoder and decoder, there are n selection lines and 2^n input lines. So, there is a total of 2^N possible combinations of inputs. A multiplexer is also treated as **Mux**.

Example: 2×1 Multiplexer

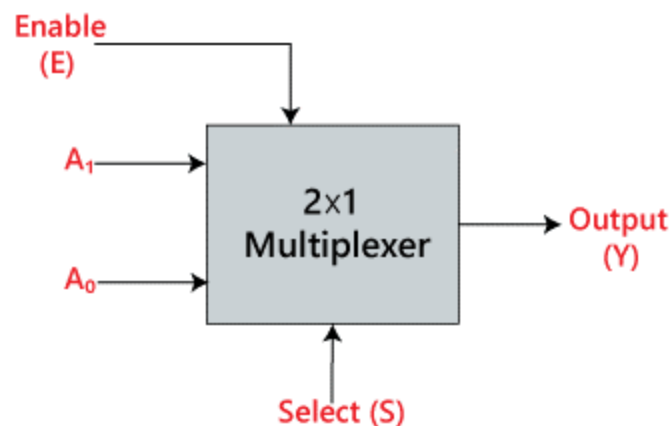


Figure 10:Block Diagram:

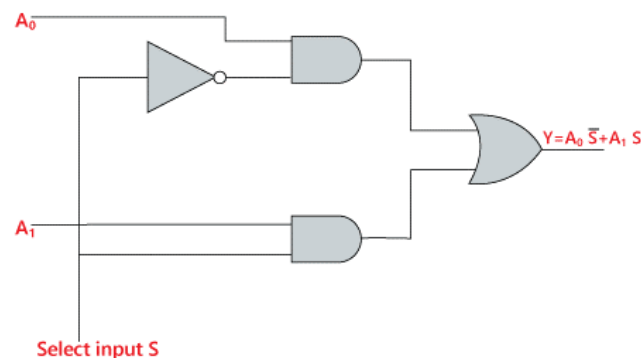


Figure 11:Logical circuit

INPUTS	Output
S_0	Y
0	A_0
1	A_1

Figure 12: Truth Table:

De-multiplexer:

A De-multiplexer is a combinational circuit that has only 1 input line and 2^N output lines. Simply, the multiplexer is a single-input and multi-output combinational circuit. The information is received from the single input lines and directed to the output line. On the basis of the values of the selection lines, the input will be connected to one of these outputs. De-multiplexer is opposite to the multiplexer.

Unlike encoder and decoder, there are n selection lines and 2^n outputs. So, there is a total of 2^n possible combinations of inputs. De-multiplexer is also treated as **De-mux**.

Example: 2×1 De-multiplexer:

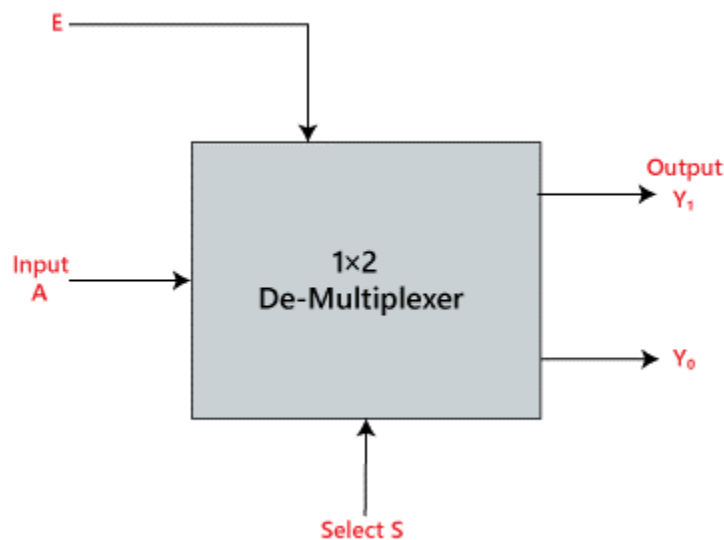


Figure 13: Block Diagram

INPUTS	Output	
S_0	Y_1	Y_0
0	0	A
1	A	0

Figure 14: Truth Table

The logical expression of the term Y is as follows:

$$Y_0 = S_0' \cdot A$$

$$Y_1 = S_0 \cdot A$$

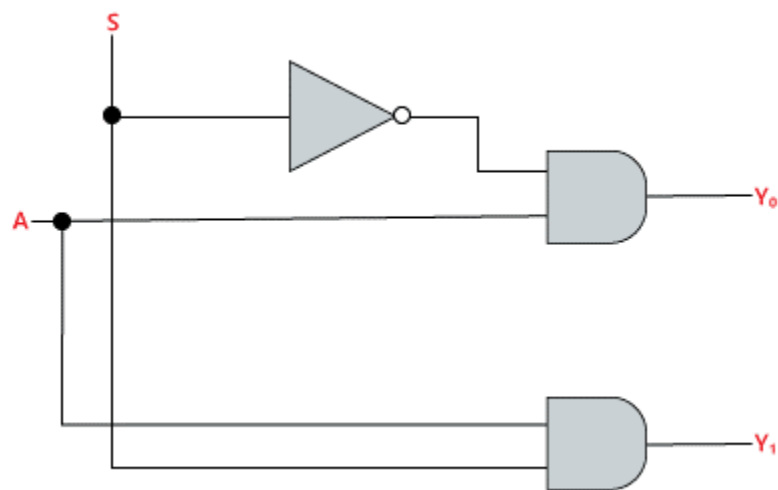


Figure 15: Logical circuit

The procedure & Data and results:

Constructing a 4-to-2 Encoder with Basic Gates

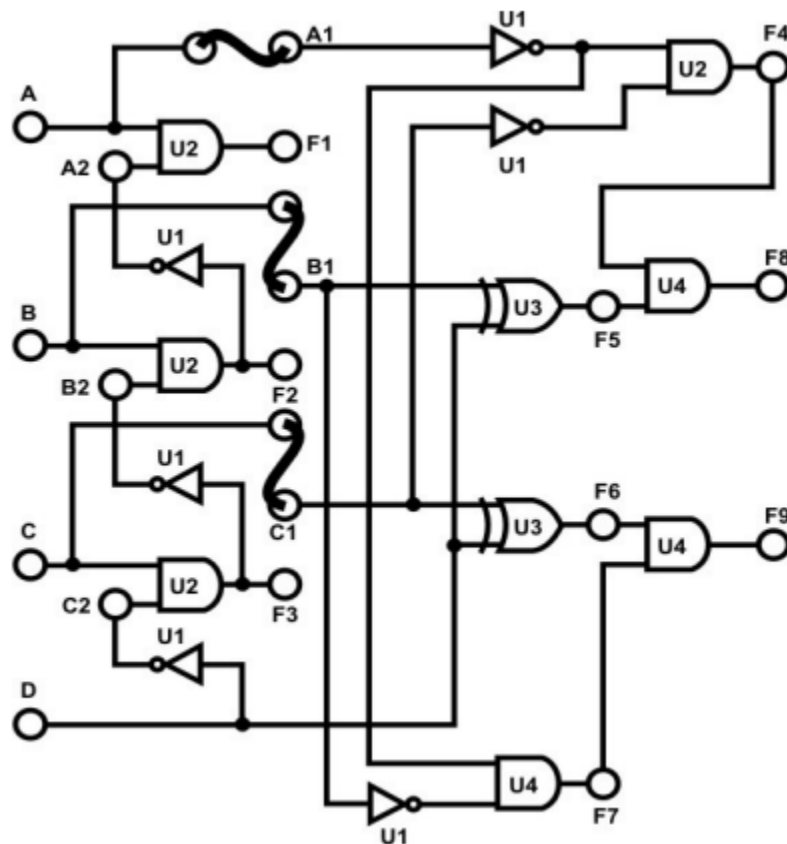


Figure 16: wiring diagram of 4-to-2-line Encoder.

D	C	B	A	F9	F8
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	1
0	0	1	1	0	0
0	1	0	0	1	0
0	1	0	1	0	0
0	1	1	0	0	0
0	1	1	1	0	0
1	0	0	0	1	1
1	0	0	1	0	0
1	0	1	0	0	0
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	0	0

Figure 17:Table 5

it can be noticed that the encoder used was an active-high input-output encoder.

Constructing 9-to-4-Line Encoder with TTL IC:

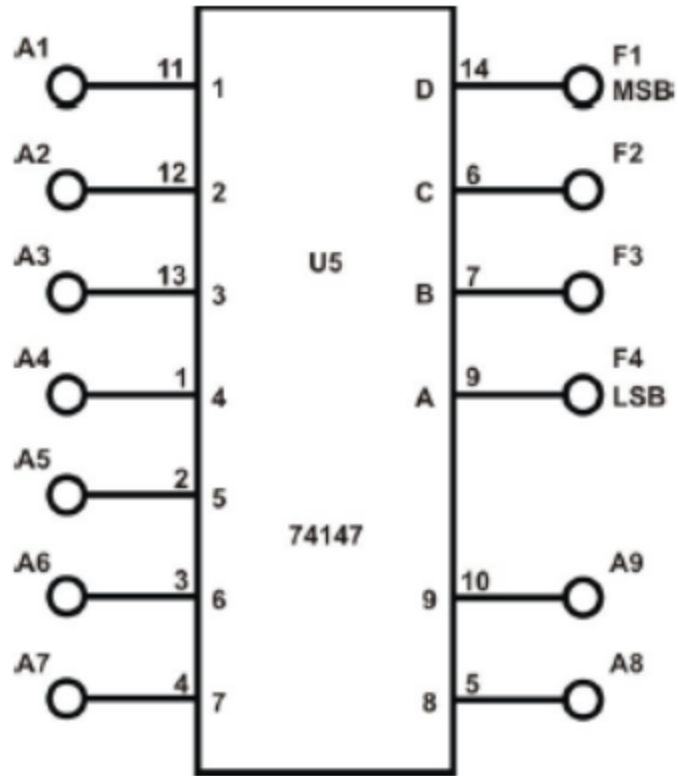


Figure 18: 74147 BCD Priority Encoder

A9	A8	A7	A6	A5	A4	A3	A2	A1	F4	F3	F2	F1
0	1	1	1	1	1	1	1	1	0	1	1	0
0	0	1	1	1	1	1	1	1	0	1	1	0
1	1	1	1	1	1	1	1	0	0	1	1	1
1	1	1	1	1	1	1	0	0	1	0	1	1
1	1	1	1	1	1	0	1	1	0	0	1	1
1	1	1	1	1	0	0	0	0	1	1	0	1
1	1	1	1	0	1	1	1	1	0	1	0	1
1	1	1	1	0	0	0	0	1	0	1	0	1
1	1	1	0	1	1	1	0	0	1	0	0	1
1	1	0	1	1	0	1	1	0	0	0	0	1
1	1	0	0	0	1	1	1	1	0	0	0	1
1	0	0	0	0	1	1	1	1	0	1	1	1

Figure 19 .Table 6

Constructing 2-to-4 Line Decoder with Basic Gates:

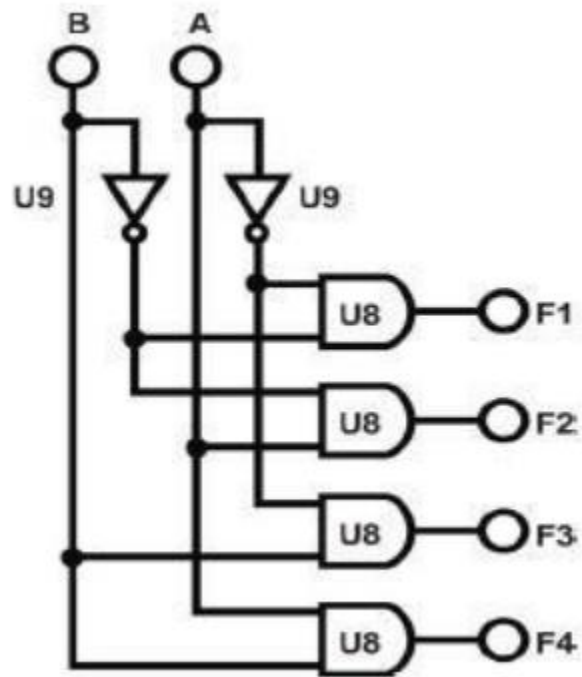


Figure 20-2 :to-4 Decoder.

B	A	F1	F2	F3	F4
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Figure 21:Table 7

It can be noticed that an active high input-output decoder was used

Constructing 4-to-10 Line Decoder with TTL IC:

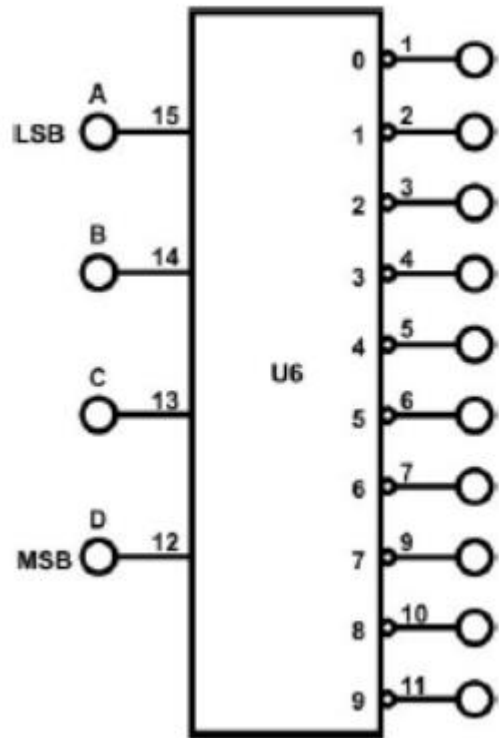


Figure 22:4-to-10-line Decoder

	Input	Input	Input	Input	Output	Output	Output	Output	Output	Output	Output	Output	Output	Output
	D	C	B	A	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	1	1	1	1	1	1	1	1	1
1	0	0	0	1	1	0	1	1	1	1	1	1	1	1
2	0	0	1	0	1	1	0	1	1	1	1	1	1	1
3	0	0	1	1	1	1	1	0	1	1	1	1	1	1
4	0	1	0	0	1	1	1	1	0	1	1	1	1	1
5	0	1	0	1	1	1	1	1	1	0	1	1	1	1
6	0	1	1	0	1	1	1	1	1	1	0	1	1	1
7	0	1	1	1	1	1	1	1	1	1	1	0	1	1
8	1	0	0	0	1	1	1	1	1	1	1	1	0	1
9	1	0	0	1	1	1	1	1	1	1	1	1	1	0

Figure 23:Table 8

We can see that an active-high input, active- low output decoder was used according to the output.

Constructing 2-to-1-Line Multiplexer with basic Gates: -

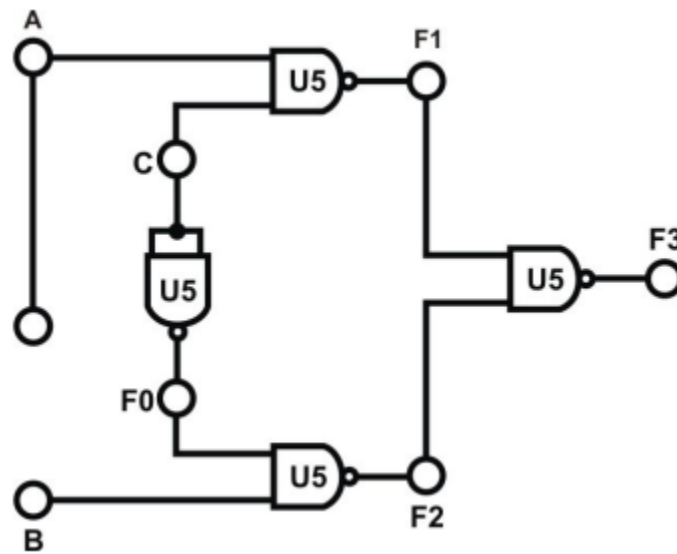


Figure 24:2-to-1 Multiplexer.

C	A	B	F3
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Figure 25:Table 9

The above table represents an active-high input-output multiplexer with C as the selection bit.

Constructing 8-to-1 Line Multiplexer with IC:

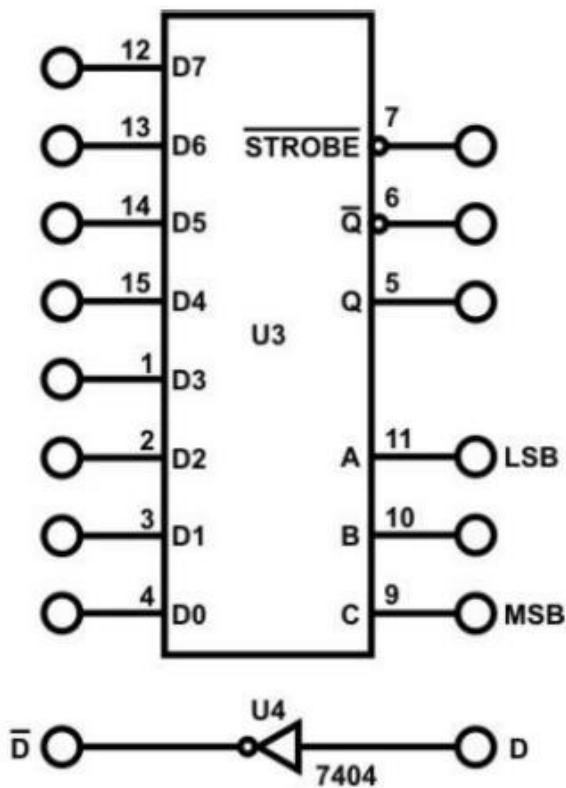


Figure 26-8 .to-1 MUX

c	B	A	Q
0	0	0	D0
0	0	1	D1
0	1	0	D2
0	1	1	D3
1	0	0	D4
1	0	1	D5
1	1	0	D6
1	1	1	D7

Figure 27:Table 10

As we see in the table IC doesn't set the output based on the inputs.

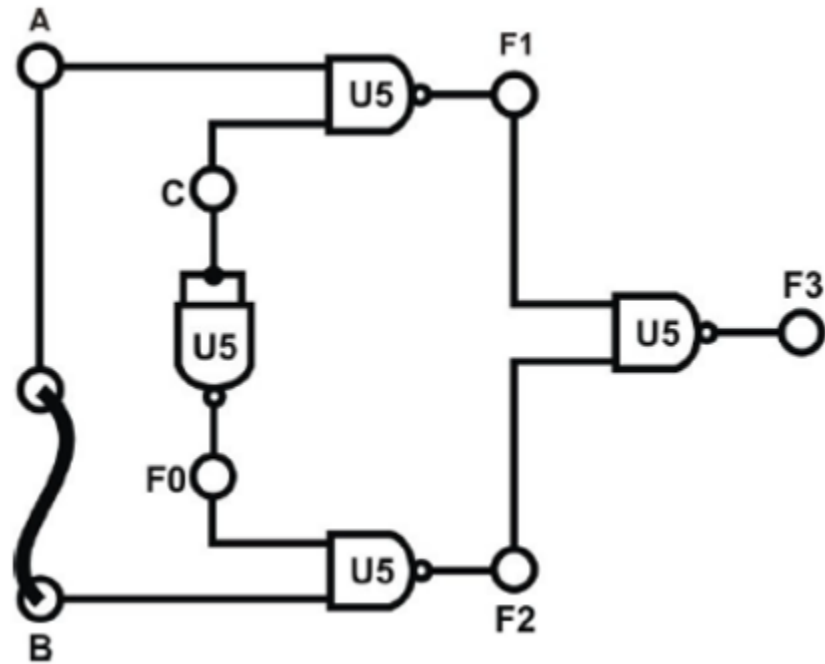
Using Multiplexer to implement a Logic Function:

Given the following function: $F(A, B, C, D) = \sum (0, 2, 4, 5, 7, 8, 10, 11, 15)$ 1. Implement the function using the 8-to-1 multiplexer using Block Multiplexer 1 of module IT-3005. Note that we have 4 inputs for this function. Three of them will be represented by the controls C B A.

A	B	c	D	y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Figure 28:Table 11

Constructing 1-to-2 Line Demultiplexer with Basic Logic Ga



C	A	F1	F2	F3
0	0	1	1	0
0	1	1	0	1
1	0	1	1	0
1	1	0	1	1

Figure 29:Table 12

The above table represents an active-high input-output demultiplexer with C as the selection bit.

Constructing 1-to-8-Line Demultiplexer with CMOS IC:

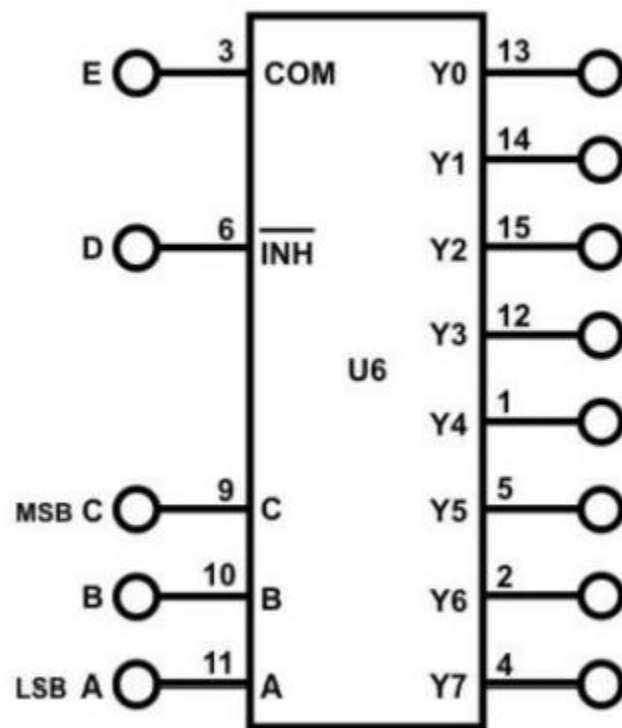


Figure 30:1-to-8 Demultiplexer

C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Figure 31:Table 13

It can be noticed that the D is an active-low enable, when it was 1 it disables the demultiplexer, and while it's 0 it enables the demultiplexer and allows it to work properly.

Conclusion: -

At the end, by the experiment we be able to understand what is the basic and the core concepts of Decoders, Encoders, Multiplexers and Demultiplexers. Also, we learnt how to implement functions by using the combination circuits Decoders, Encoders, Multiplexers and Demultiplexers. And by using the kits we proved our knowledge practically in best way.

References:

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<https://www.javatpoint.com/de-multiplexer-digital-electronics>